What Makes Reactomics an Omics: Reactions Are the Analytes

Network construction alone is not enough — quantification at the reaction layer is the decisive shift

1. Peak list

Untargeted LC-MS
m/z + intensity
(no IDs needed)

2. PMD pairs

Match Δm to curated
reactions (O, CH_2, ...)
across all peak pairs

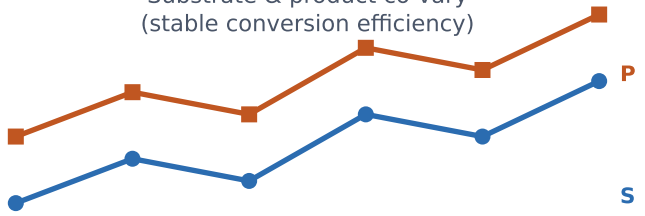
3. Reaction matrix

getreact(): reaction × sample
activity scores for stats
& comparison

Two regulation regimes → two quantitative readouts

Static PMD

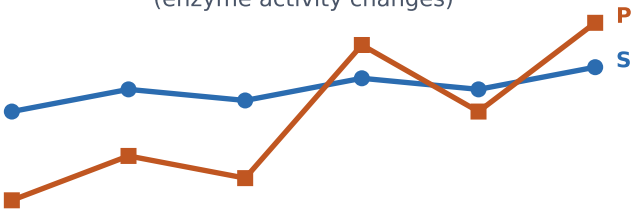
Substrate & product co-vary
(stable conversion efficiency)



Quantify: intensity sum (S + P)
⇒ upstream / downstream control

Dynamic PMD

Substrate & product diverge
(enzyme activity changes)



Quantify: ratio (stable / variable)
⇒ enzyme-level control